

PenguinDrive

An envelope filter, phaser, drive and squash compressor in one box, after a 70s art-rock live guitar rig.



Overview

PenguinDrive chains four effects that a 70s art-rock guitarist might have run on stage: a resonant envelope filter, a four-stage phaser, a valve-style drive and a compressor at the end that squashes it all together.

In ENVELOPE mode the filter follows how hard you play: every pick attack opens it upwards from the PEAK setting and it falls back as the note dies, for the vocal, quacking funk sound. In FIXED mode the filter stays put at PEAK, like a wah pedal left half-way down. The filter is a band-pass, so the effect side is focused and nasal; MIX blends it with your dry signal.

A DRIFT dial adds a slow random wander to the filter frequency and the drive, so the sound breathes a little like real analogue gear.

Reach for it for

- Funky, quacking guitar and clav lines
- Vocal, wah-like lead tones
- Driven, phased guitar for 70s rock
- Synth bass and keys that need movement and bite
- Cocked-wah tones with FIXED mode

Controls

Mode (header)

ENVELOPE	The filter follows your playing: the harder the note, the further it opens above PEAK. The yellow light above the key shows it is on.
FIXED	The filter sits still at PEAK - a cocked-wah tone. SENS has no effect. The orange light above the key shows it is on.
OFF	Bypasses the filter, phaser, drive and squash. The signal still passes through the input and output stages.

Display

The window on the left shows the current filter frequency (FILTER PEAK) in Hz, with a moving trace that follows it.

Main controls

The signal runs through the controls in this order: filter, phaser, drive, squash.

Z-DRIVE	How hard the filtered signal drives the saturation stage, from 0 to 10. 0 is almost clean; 10 adds up to 25 dB of gain into the soft clipper for a thick, compressed drive.
SENS	Envelope sensitivity, 0 to 100%: how far your playing pushes the filter up from PEAK. Low settings give a subtle movement; high settings sweep it right up for each note. ENVELOPE mode only.
PEAK	The filter's resting frequency, 200 Hz to 5 kHz. In ENVELOPE mode this is where the sweep starts; in FIXED mode it is where the filter stays.
Q / RES	The filter's resonance, 10 to 95%. Higher settings narrow the band and make it ring, for a more vocal, quacking sound.
PHASER	How much of the four-stage phaser is blended in after the filter, 0 to 100%. At 0 the phaser is off. It sweeps slowly

between about 500 Hz and 2 kHz.

SQUASH

The compressor at the end of the chain, 0 to 100%. Higher settings lower its threshold for a flatter, more sustained sound; it grabs quickly and lets go slowly. At 0 it is off.

Chaos drift

DRIFT

0 to 100%. 50% is the standard amount; 100% is twice that, 0 turns it off. It makes the filter frequency wander slowly and the drive amount breathe a little. Higher settings also make the wander move faster.

IN / OUT

TRIM

Input level before anything else, from -24 to +12 dB.

GUARD

Click the TRIM knob (without turning it) to switch GUARD on or off; the small light at the knob's upper right shows it is on and glows brighter while it works. On by default. Gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS, so loud sources never overdrive the plugin. It never boosts quiet sources.

SAT

Band saturation on the way out: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.

OFF / CLIP / LIMIT

The final stage: nothing, a soft clipper that holds the peaks just under 0 dBFS (no latency), or a lookahead limiter (3 ms latency, compensated by REAPER). The CLIP key flashes red while the clipper is working.

Master

MIX

Dry / effect balance, 0 to 100% (50% by default). Lower settings keep more of your clean attack under the filtered, driven sound.

LOW CUT	A high-pass filter on the effected signal only, from off (fully left) up to 500 Hz. The dry low end stays as it is.
OUTPUT	Output level, -24 to +12 dB.
LEVEL	Output meters, left and right.

The header

PRESETS, ◀ ▶ and the display	PRESETS (or a click on the display) opens the list of factory presets; ◀ and ▶ step through them. A preset sets the mode and the main controls (including the phaser speed) and MIX; drift, the input and output stages, LOW CUT and OUTPUT are left as they are.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Art-Rock Lead ('73)	Envelope filter with strong resonance, a fair amount of drive, phaser and squash, mostly wet (85% mix).
Funk Envelope	High sensitivity from a low 800 Hz peak with lots of resonance and heavy squash: the classic quack.
Cocked Wah Drive	FIXED filter at 1.4 kHz, heavy drive and a faster phaser, fully wet.
Mild Auto-Sweep	Gentle envelope sweep from 2 kHz with light drive and no phaser, half wet.

Tips

- Set PEAK first with SENS at 0 to find the tone the filter rests on, then bring SENS up until the sweep follows your playing the way you want.
- The envelope responds to level, so a compressor or a louder TRIM before the filter makes the sweep more even; play dynamically for a more expressive one.
- The phaser's speed has no knob on the panel: the presets set it (0.1 to 0.5 Hz), and it can be changed in REAPER's parameter list as Phaser Speed (0.05 to 5 Hz).
- For a cleaner funk sound, keep Z-DRIVE low and use SQUASH for sustain instead.

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